

Viral Infections & Vaccinations in CKD

Nephrology Patient Education Series · williamriveromd.com

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Uremic Immunoparalysis: 5 Simultaneous Failures

Immune Arm	What Goes Wrong in CKD/Dialysis
T-cell function	Exhausted, reduced proliferation, impaired cytokine response
B-cell response	Poor activation, reduced antibody production, faster waning
NK cell activity	Decreased cytotoxicity against viral-infected cells
Neutrophil function	Impaired phagocytosis, delayed recruitment to infection site
Complement system	Low C3/C4, impaired opsonization, missed pathogen recognition

■ **Clinical Alert**
Viral infections that clear in 5–7 days in healthy adults can trigger **AKI, CKD progression, or death** in dialysis patients.

Seroconversion Rates by CKD Stage

Population	Vaccine Seroconversion Rate
Healthy adult	~95%
CKD Stage 3	~80%
CKD Stage 4–5	~60%
Hemodialysis	50–60%
Peritoneal Dialysis	65–75%

Key Message: Vaccinate BEFORE dialysis. Every month of delay = lost immunogenicity.

Hepatitis B: The Highest-Stakes Vaccine in CKD

Dose	Standard Adult	CKD / Dialysis
Antigen	10 mcg	40 mcg
Schedule	3 doses (0/1/6 mo)	4 doses (0/1/2/6 mo)
Brand example	Engerix-B 10 mcg	Engerix-B 40 mcg or HEPLISAV-B
Response rate	~95% seroconvert	50–70% with standard; 70–85% with double-dose

Monitoring Protocol (Anti-HBs Titer Targets)

Timing	Check anti-HBs 4–8 weeks after completing the full series
Target	≥ 100 IU/L (NOT just ≥10 IU/L — the CKD standard)
50–99 IU/L	Give a single booster dose; recheck anti-HBs in 4–8 weeks
<10 IU/L	Repeat full 4-dose series; consider switching to HEPLISAV-B (adjuvanted)
Dialysis	Check anti-HBs annually ; re-boost if titer falls below 100 IU/L

HBV Risk in Dialysis Patients

Risk Factor	Impact
HBV prevalence	Up to 10× higher than general population in dialysis units
Non-response rate (standard dose)	30–50% in dialysis vs ~5% in healthy adults
Main risk factors	Loss of uremic immune function + inadequate standard dosing

Standard of Care for CKD: Double dose (40 mcg) + 4-dose schedule + anti-HBs target ≥100 IU/L = the minimum acceptable approach for all CKD patients.

Influenza, COVID-19 & Pneumococcal Vaccines

Influenza (IIV — Inactivated Influenza Vaccine)

Parameter	Recommendation
Frequency	Annual vaccination — every year without exception
Timing (PH)	Give BEFORE rainy/cold season: July–November Philippines
Preferred formulation	High-dose (Fluzone HD) or adjuvanted (FLUAD) for CKD Stage 4–5 and dialysis if available
Household cocooning	ALL household contacts should receive annual influenza vaccine to protect the CKD patient
Contraindication	NO live attenuated influenza vaccine (LAIV / nasal spray) in CKD

COVID-19 Vaccines

Parameter	Recommendation
Primary series	2 doses mRNA (Pfizer BNT162b2 or Moderna mRNA-1273) OR 2 doses viral vector
Immunocompromised	3-dose primary series for dialysis patients and those on immunosuppression
Booster	Recommended every 12 months for dialysis and transplant patients
Timing with other vaccines	Can give with other non-live vaccines on the same day

Pneumococcal Vaccine

Parameter	Recommendation
Preferred approach	PCV20 alone (preferred) — covers 20 serotypes; NO PPSV23 needed if PCV20 was used. If PCV15 used instead: add PPSV23 ≥8 weeks later. If only PPSV23 available: give PPSV23 alone (less immunogenic).
Who needs it	Both pre-dialysis AND dialysis patients; also transplant candidates (pre-Tx)
PPSV23 re-doses	Only if PPSV23 was used (not PCV20). CKD and dialysis: 2 PPSV23 doses total — ≥5 years apart. Dose 1 at diagnosis, Dose 2 ≥5 years later. At age 65+ with ≥5 years since last: one additional dose (max 3 lifetime). If PCV20 was used: no re-vaccination needed.

Remember: CKD patients are at high risk for pneumococcal pneumonia, invasive pneumococcal disease, severe influenza, and complicated COVID-19. These three vaccines together represent the core annual vaccination program for every CKD patient.

Other Important Vaccines for CKD Patients

Varicella-Zoster Virus (VZV) & Shingrix

Vaccine	CKD Guidance
Live Varicella (Varivax)	CONTRAINDICATED in dialysis and CKD Stage 4–5
Live Varicella (CKD 1–3)	May give if VZV-seronegative AND not on immunosuppression; avoid if ≥Stage 4
Shingrix (recombinant, non-live)	PREFERRED over Zostavax — 2 doses, 2–6 months apart
Shingrix indications	All CKD patients ≥50 years OR any age on immunosuppression (any CKD stage)

Hepatitis A

Parameter	Recommendation
Schedule	2-dose series at 0 and 6–12 months
Dose	Standard adult dose; no CKD-specific adjustment
Pre-screening	Check anti-HAV IgG first — many Filipinos naturally immune from childhood
If seropositive	No vaccine needed; document immunity

HPV (Gardasil 9)

Parameter	Recommendation
Schedule	3-dose series (0 / 2 / 6 months) for CKD patients regardless of age criteria
Importance	Particularly important for pre-transplant patients to prevent HPV-related cancers
Post-Tx	Complete series pre-transplant if possible; limited data post-transplant

Tdap / Td (Tetanus, Diphtheria, Pertussis)

Parameter	Recommendation
Initial booster	Single Tdap if >10 years since last tetanus-containing vaccine
Subsequent	Td booster every 10 years thereafter
No CKD adjustment	Standard dose; give regardless of CKD stage or dialysis status

Meningococcal Vaccine

Parameter	Recommendation
MenACWY (2 doses)	For asplenic patients, transplant candidates, or those in endemic/high-density settings
MenB (recombinant)	If additional risk factors present (complement deficiency, properdin deficiency)
Schedule	2 doses MenACWY 8 weeks apart for high-risk; booster every 5 years if risk persists

Philippine Vaccination Schedule for CKD Patients

Reference table — bring to every clinic visit. All vaccines should be confirmed available through your local DOH or hospital pharmacy.

Vaccine	CKD Stage 1–3	CKD Stage 4–5	Hemodialysis	Peritoneal Dialysis	Post-Transplant
Hepatitis B	Standard 3×10 mcg	Double-dose 4×40 mcg	Double-dose 4×40 mcg	Double-dose 4×40 mcg	No (if pre-Tx done); boost if anti-HBs <100
Influenza (IIV)	Annual	Annual (high-dose preferred)	Annual (high-dose preferred)	Annual	Annual
COVID-19 mRNA	Primary + annual booster	Primary + annual booster	3-dose primary + annual booster	3-dose primary + annual booster	Delay 3–6 mo post-Tx
Pneumococcal (PCV20 alone, or PCV15+PPSV23)	PCV20 alone (preferred) OR PCV15 → PPSV23 ≥8 wk later	PCV20 alone (preferred) OR PCV15 → PPSV23 ≥8 wk later	PCV20 alone (preferred). If PPSV23 used: 2nd dose ≥5 yr later (max 2 doses <65, 3 lifetime)	PCV20 alone (preferred). If PPSV23 used: 2nd dose ≥5 yr later (max 2 doses <65, 3 lifetime)	Give pre-Tx; defer 3–6 mo post-Tx if missed
Shingrix	≥50 yr or immunosuppressed	Yes (all CKD 4–5)	Yes (all HD patients)	Yes (all PD patients)	Delay 6–12 mo post-Tx
Hepatitis A	2 doses if sero-neg	2 doses if sero-neg	2 doses if sero-neg	2 doses if sero-neg	No (pre-Tx only)
HPV (Gardasil 9)	3 doses if eligible	3 doses	3 doses	3 doses	Pre-Tx only
Tdap	1 booster	1 booster	1 booster	1 booster	3–6 mo post-Tx
Varicella / live VZV	CKD 1–3 only if sero-neg	AVOID	AVOID	AVOID	AVOID
MMR (live)	CKD 1–3 only	AVOID	AVOID	AVOID	AVOID (pre-Tx ≥4 wk before)
Meningococcal	If indicated	If indicated	If indicated	If indicated	Yes if indicated

All vaccines should be confirmed available through your local DOH/hospital pharmacy. Contact Dr. Rivero's clinic for individualized schedules.

Live vs. Non-Live Vaccines: Safety Rules in CKD

■ LIVE VACCINES — Use with Caution / Avoid

- MMR (measles-mumps-rubella)
- Varicella / Chickenpox (Varivax)
- Zostavax (old shingles — use Shingrix instead)
- LAIV (nasal flu spray)
- Yellow fever
- Oral typhoid (Ty21a)

Rules:

- **AVOID** in CKD Stage 4–5, dialysis, and post-transplant
- CKD Stage 1–3 (non-immunosuppressed): give if indicated
- Must be given ≥4 weeks before transplant if applicable

✓ NON-LIVE VACCINES — Safe in CKD

- Hepatitis B (Engerix-B, HEPLISAV-B)
- Influenza IIV (injectable only)
- COVID-19 mRNA (Pfizer, Moderna)
- PCV20 / PPSV23
- Hepatitis A
- Shingrix (recombinant zoster)
- Gardasil 9 (HPV)
- Tdap / Td
- Meningococcal (MenACWY, MenB)

Rules:

- Safe across **ALL** CKD stages including dialysis
- Safe post-transplant (timing may vary by stage)
- Prefer non-live alternatives whenever available

Post-Transplant Vaccination Timeline

0–3 months	NO vaccines — highest immunosuppression (mycophenolate + tacrolimus + steroids)
3–6 months	COVID booster, annual influenza; check anti-HBs titer
6–12 months	Tdap, Hepatitis A (if needed), HPV, Meningococcal
>12 months	Shingrix, PCV20; resume normal non-live vaccine schedule
NEVER post-transplant	MMR, varicella, LAIV, Zostavax, yellow fever, oral typhoid

My Personal Vaccination Record

CKD / Dialysis Patient — Fill in using Adobe Reader, Preview, or browser PDF viewer, then print

FULL NAME	DATE OF BIRTH	AGE	SEX	LOG DATE
CKD STAGE / MODALITY	NEPHROLOGIST			CLINIC

Vaccine	Date Given	Dose #	Lot / Batch #	Given By / Clinic	Next Due / Booster	Notes / Anti
Hepatitis B — Dose 1 (40 mcg)						
Hepatitis B — Dose 2 (40 mcg)						
Hepatitis B — Dose 3 (40 mcg)						
Hepatitis B — Dose 4 (40 mcg)						
Influenza IIV (Season: _____)						
Influenza IIV (Season: _____)						
COVID-19 — Primary Dose 1						
COVID-19 — Primary Dose 2						
COVID-19 — Booster 1						
COVID-19 — Booster 2						
Pneumococcal PCV20						
PPSV23 — Dose 1						
PPSV23 — Dose 2 (≥5 yrs)						
Shingrix (Zoster) — Dose 1						
Shingrix (Zoster) — Dose 2						
Hepatitis A — Dose 1						
Hepatitis A — Dose 2						
Tdap / Td						
HPV (Gardasil 9) — Dose 1						
HPV (Gardasil 9) — Dose 2						
HPV (Gardasil 9) — Dose 3						
Other: _____						
Other: _____						

Anti-HBs target for CKD/dialysis patients: ≥100 IU/L. Check annually in dialysis patients. Max 3 lifetime PPSV23 doses. Confirm your vaccination plan with